

Earley Parser Assignment Report

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March 27, 2026

GitHub Repository: [Link to Code](#)

Question 1: Earley Algorithm Chart

Task: Emulate the behaviour of the Earley algorithm on the sentence "*time flies like an arrow*" using the grammar rules provided in Figure 1.

Below is the final chart generated by the parser. Completed rules are marked with an asterisk (*).

```
=====
EARLEY CHART
=====

-- Chart[0] (before reading any word) -----
[0,0] S -> . NP VP
[0,0] NP -> . N
[0,0] NP -> . Det N
[0,0] NP -> . N N
[0,0] N -> . time
[0,0] N -> . flies
[0,0] N -> . arrow
[0,0] Det -> . a
[0,0] Det -> . an

-- Chart[1] (after reading 'time') -----
* [0,1] N -> time .
* [0,1] NP -> N .
  [0,1] NP -> N . N
  [0,1] S -> NP . VP
  [1,1] N -> . time
  [1,1] N -> . flies
  [1,1] N -> . arrow
  [1,1] VP -> . V NP
  [1,1] VP -> . V ADVP
  [1,1] V -> . flies
  [1,1] V -> . like

-- Chart[2] (after reading 'flies') -----
* [1,2] N -> flies .
* [1,2] V -> flies .
* [0,2] NP -> N N .
  [1,2] VP -> V . NP
  [1,2] VP -> V . ADVP
  [0,2] S -> NP . VP
  [2,2] NP -> . N
```

```

[2,2] NP -> . Det N
[2,2] NP -> . N N
[2,2] ADVP -> . ADV NP
[2,2] VP -> . V NP
[2,2] VP -> . V ADVP
[2,2] N -> . time
[2,2] N -> . flies
[2,2] N -> . arrow
[2,2] Det -> . a
[2,2] Det -> . an
[2,2] ADV -> . like
[2,2] V -> . flies
[2,2] V -> . like

-- Chart[3] (after reading 'like') -----
* [2,3] ADV -> like .
* [2,3] V -> like .
  [2,3] ADVP -> ADV . NP
  [2,3] VP -> V . NP
  [2,3] VP -> V . ADVP
  [3,3] NP -> . N
  [3,3] NP -> . Det N
  [3,3] NP -> . N N
  [3,3] ADVP -> . ADV NP
  [3,3] N -> . time
  [3,3] N -> . flies
  [3,3] N -> . arrow
  [3,3] Det -> . a
  [3,3] Det -> . an
  [3,3] ADV -> . like

-- Chart[4] (after reading 'an') -----
* [3,4] Det -> an .
  [3,4] NP -> Det . N
  [4,4] N -> . time
  [4,4] N -> . flies
  [4,4] N -> . arrow

-- Chart[5] (after reading 'arrow') -----
* [4,5] N -> arrow .
* [3,5] NP -> Det N .
* [2,5] ADVP -> ADV NP .
* [2,5] VP -> V NP .
* [1,5] VP -> V ADVP .
* [0,5] S -> NP VP .

```

===== PARSE TREES =====

```

-- Tree 1 (Verb Phrase "flies") -----
Probability : 0.0044800000
Cost (-log2): 7.8022855523
(S
  (NP (N time))
  (VP (V flies)
    (ADVP (ADV like)
      (NP (Det an) (N arrow))))))

-- Tree 2 (Noun Phrase "time flies") -----
Probability : 0.0009600000
Cost (-log2): 10.0246779737
(S
  (NP (N time) (N flies))
  (VP (V like)
    (NP (Det an) (N arrow))))

```

Question 2: Compute Tree Probabilities

Task: Compute the probability of each tree emitted by the parser using the rule probabilities from Figure 1.

The sentence yields two distinct parse trees. The probability of each tree is the product of the probabilities of all rules used in its derivation. The cost is calculated using the negative base-2 logarithm ($-\log_2$) of that probability.

Tree 1 (Noun Phrase "time flies"):

- **Derivation:** $[\text{time flies}]_{\text{NP}} [\text{like } [\text{an arrow}]_{\text{NP}}]_{\text{VP}}$
- **Probability Calculation:** $P(S \rightarrow \text{NP VP}) \times P(\text{NP} \rightarrow \text{N N}) \times P(\text{N} \rightarrow \text{time}) \times P(\text{N} \rightarrow \text{flies}) \times P(\text{VP} \rightarrow \text{V NP}) \times P(\text{V} \rightarrow \text{like}) \times P(\text{NP} \rightarrow \text{D N}) \times P(\text{D} \rightarrow \text{an}) \times P(\text{N} \rightarrow \text{arrow})$
- **Values:** $1.0 \times 0.25 \times 0.4 \times 0.2 \times 0.6 \times 0.5 \times 0.4 \times 1.0 \times 0.4 = \mathbf{0.00096}$
- **Log Probability (Cost):** $-\log_2(0.00096) \approx \mathbf{10.02468}$

Tree 2 (Verb Phrase "flies"):

- **Derivation:** $[\text{time}]_{\text{NP}} [\text{flies } [\text{like } [\text{an arrow}]_{\text{NP}}]_{\text{ADVP}}]_{\text{VP}}$
- **Probability Calculation:** $P(S \rightarrow \text{NP VP}) \times P(\text{NP} \rightarrow \text{N}) \times P(\text{N} \rightarrow \text{time}) \times P(\text{VP} \rightarrow \text{V ADVP}) \times P(\text{V} \rightarrow \text{flies}) \times P(\text{ADVP} \rightarrow \text{ADV NP}) \times P(\text{ADV} \rightarrow \text{like}) \times P(\text{NP} \rightarrow \text{D N}) \times P(\text{D} \rightarrow \text{an}) \times P(\text{N} \rightarrow \text{arrow})$
- **Values:** $1.0 \times 0.35 \times 0.4 \times 0.4 \times 0.5 \times 1.0 \times 1.0 \times 0.4 \times 1.0 \times 0.4 = \mathbf{0.00448}$
- **Log Probability (Cost):** $-\log_2(0.00448) \approx \mathbf{7.80229}$

Tree 2 is approximately 4.67 times more probable under the provided grammar, requiring fewer bits to encode (lower cost).

Question 3: Grammar Design for Ambiguity

Task: Design a grammar to give analyses for *"the man shot the soldier with a gun"* and test the parser implementation.

Designed Grammar

To capture the Prepositional Phrase (PP) attachment ambiguity (whether the man shot using the gun, or the soldier possessed the gun), the following grammar was designed with equal probabilities (0.5) for structural branching:

```

1.0   S    NP VP
0.5   NP   DT N
0.5   NP   DT N PP
0.5   VP   V NP
0.5   VP   V NP PP
1.0   PP   P NP
0.5   DT   the
0.5   DT   a
0.25  N    man
0.25  N    soldier
0.25  N    gun
1.0   V    shot
1.0   P    with

```

Final Chart & Parse Trees

The parser successfully emitted two valid analyses with identical probabilities (0.000122), reflecting the ambiguity of the PP attachment.

```

=====
EARLEY CHART
=====

-- Chart[0] (before reading any word) -----
[0,0] S -> . NP VP
[0,0] NP -> . DT N
[0,0] NP -> . DT N PP
[0,0] DT -> . the
[0,0] DT -> . a

-- Chart[1] (after reading 'the') -----
* [0,1] DT -> the .
  [0,1] NP -> DT . N
  [0,1] NP -> DT . N PP
  [1,1] N -> . man
  [1,1] N -> . soldier
  [1,1] N -> . gun
  [1,1] N -> . arrow

-- Chart[2] (after reading 'man') -----
* [1,2] N -> man .
* [0,2] NP -> DT N .
  [0,2] NP -> DT N . PP
  [0,2] S -> NP . VP
  [2,2] PP -> . P NP
  [2,2] VP -> . V NP
  [2,2] VP -> . V NP PP
  [2,2] P -> . with
  [2,2] V -> . shot

-- Chart[3] (after reading 'shot') -----
* [2,3] V -> shot .
  [2,3] VP -> V . NP
  [2,3] VP -> V . NP PP
  [3,3] NP -> . DT N
  [3,3] NP -> . DT N PP
  [3,3] DT -> . the
  [3,3] DT -> . a

```

```

-- Chart[4] (after reading 'the') -----
* [3,4] DT -> the .
  [3,4] NP -> DT . N
  [3,4] NP -> DT . N PP
  [4,4] N -> . man
  [4,4] N -> . soldier
  [4,4] N -> . gun
  [4,4] N -> . arrow

-- Chart[5] (after reading 'soldier') -----
* [4,5] N -> soldier .
* [3,5] NP -> DT N .
  [3,5] NP -> DT N . PP
* [2,5] VP -> V NP .
  [2,5] VP -> V NP . PP
  [5,5] PP -> . P NP
* [0,5] S -> NP VP .
  [5,5] P -> . with

-- Chart[6] (after reading 'with') -----
* [5,6] P -> with .
  [5,6] PP -> P . NP
  [6,6] NP -> . DT N
  [6,6] NP -> . DT N PP
  [6,6] DT -> . the
  [6,6] DT -> . a

-- Chart[7] (after reading 'a') -----
* [6,7] DT -> a .
  [6,7] NP -> DT . N
  [6,7] NP -> DT . N PP
  [7,7] N -> . man
  [7,7] N -> . soldier
  [7,7] N -> . gun
  [7,7] N -> . arrow

-- Chart[8] (after reading 'gun') -----
* [7,8] N -> gun .
* [6,8] NP -> DT N .
  [6,8] NP -> DT N . PP
* [5,8] PP -> P NP .
  [8,8] PP -> . P NP
* [3,8] NP -> DT N PP .
* [2,8] VP -> V NP PP .
  [8,8] P -> . with
* [2,8] VP -> V NP .
  [2,8] VP -> V NP . PP
* [0,8] S -> NP VP .

```

===== PARSE TREES =====

```

-- Tree 1 -----
Probability : 0.0001220711
Cost (-log2): 13.000000

(S
  (NP
    (DT the)
    (N man))
  (VP
    (V shot)
    (NP
      (DT the)
      (N soldier))
    (PP
      (P with)

```

```

      (NP
      (DT a)
      (N gun))))))

-- Tree 2 -----
Probability : 0.0001220711
Cost (-log2): 13.000000

(S
  (NP
    (DT the)
    (N man))
  (VP
    (V shot)
    (NP
      (DT the)
      (N soldier)
      (PP
        (P with)
        (NP
          (DT a)
          (N gun)))))))

```

Question 4: Implementation & Efficiency

Task: Explain briefly how the submitted code solved the problems of Correctness and Efficiency.

Correctness

Every chart entry `chart[col][key]` stores a dictionary containing two keys: `'cost'` (the lowest negative base-2 log-probability) and `'backs'` (a list of all backpointers).

Tracking best derivation: When a new item is inserted, its base-2 log cost is stored. If the same item arrives again via a different derivation, the new cost is compared against the stored `'cost'`. If it is **lower**, `'cost'` is updated. This guarantees the parser maintains the Viterbi minimum cost.

Reacting to new derivations: The new backpointer is appended to the `'backs'` list if it is not already present, regardless of whether its cost was lower. This decouples Viterbi cost tracking from tree structure tracking, allowing the post-parse step to exhaustively enumerate all distinct parse trees.

Efficiency

The algorithm strictly ensures $O(n^2)$ space and $O(n^3)$ time complexities:

Space $O(n^2)$: The chart contains $n + 1$ columns. Each item key is a tuple (`rule_idx`, `dot`, `origin`). For a fixed grammar G , `rule_idx` and `dot` are constants. The `origin` index ranges from 0 to j (at most $n + 1$ values per column). Thus, each column holds at most $O(|G| \cdot n)$ items, yielding a total space complexity of $O(n^2)$.

Time $O(n^3)$: The PREDICT and SCAN operations require $O(|G|)$ and $O(1)$ work respectively. The dominant operation is COMPLETE. For each complete item starting at `origin` i and ending at j , it scans `chart[i]` for items waiting for the completed non-terminal. Since `chart[i]` has at most $O(|G| \cdot n)$ items, and there are $O(n^2)$ possible (i, j) pairs, the total time complexity is bounded by $O(n^3)$.

$O(1)$ Push & Duplicate Check: Each column `chart[col]` is implemented as a Python dictionary (hash table). Therefore, checking `key not in chart[col]` and inserting new items runs in amortized $O(1)$ time. This constant-time check is vital; without it, items could be added to the agenda multiple times, triggering redundant COMPLETE loops and leading to exponential time blowup.